

=====

KIRLOSKAR BROTHERS LIMITED
OPERATION & MAINTENANCE MANUAL
Model: KDS-450 Centrifugal Pump Series
Document Ref: OEM-KBL-KDS450-001
Issue: Rev 3.1 | Date: March 2020

=====

(EXCERPT – Sections 6, 7, and 9 relevant to AssetIQ demo)

=====

SECTION 6 – LUBRICATION

6.1 BEARING LUBRICATION TYPE

The KDS-450 series uses grease-lubricated antifriction (rolling element) bearings. Approved grease type: Lithium complex grease, NLGI Grade 2. Recommended grade: SKF LGEP 2 or equivalent.

DO NOT mix different grease types. Mixing can cause thickener incompatibility and accelerated bearing failure.

6.2 GREASE QUANTITY

Drive End (DE) Bearing : 35 grams per lubrication cycle
Non-Drive End (NDE) Bearing: 25 grams per lubrication cycle

Over-greasing is as harmful as under-greasing. Excess grease causes churning, heat generation, and premature bearing failure. Use a calibrated grease gun to ensure correct quantity.

6.3 LUBRICATION INTERVAL

Lubrication frequency depends on operating conditions:

Condition A – Standard operation (<40°C ambient, <3000 hrs/year):
Lubrication every 45 days

Condition B – Elevated temperature (40-50°C ambient) or continuous operation (>3000 hrs/year):
Lubrication every 30 days

Condition C – High temperature (>50°C ambient) or dusty/humid environment:
Lubrication every 15 days

IMPORTANT: For pumps operating in Indian coastal or monsoon environments, Condition B or C applies. Consult Kirloskar regional service team if in doubt.

6.4 BEARING TEMPERATURE LIMITS

Normal Operating Range : 40°C to 65°C above ambient
Advisory Limit : 70°C (investigate cause if sustained)
Warning Alarm : 75°C (reduce load or inspect)
Maximum Trip Limit : 85°C (immediate shutdown mandatory)

NOTE: A sudden rise in bearing temperature of >10°C without change in operating conditions is a strong indicator of lubrication failure or bearing damage. Stop pump and investigate.

SECTION 7 – MAINTENANCE SCHEDULE

7.1 DAILY CHECKS (Operator)

- [] Check suction and discharge pressure gauges
- [] Check bearing temperature (if thermometer fitted)

- [] Listen for abnormal noise (grinding, knocking)
- [] Check gland/seal leakage (max 30 drops/minute for gland packing; zero for mechanical seal except quench flow)
- [] Check motor current from MCC

7.2 WEEKLY CHECKS (Technician)

- [] Manual vibration check with vibration meter
 - Acceptable limit: 2.8 mm/s (ISO 10816 Zone A)
 - Action required: >4.5 mm/s (ISO 10816 Zone C)
- [] Check coupling condition (if coupling guard allows)
- [] Check baseplate for any cracks or loose foundation bolts

7.3 MONTHLY (Technician + Supervisor sign-off)

- [] Complete lubrication (Section 6.1 to 6.3)
- [] Check seal quench flow and quench water quality
- [] Measure and record vibration (both DE and NDE)
- [] Record bearing temperatures under load
- [] Check all instrumentation – pressure, temperature, flow

7.4 QUARTERLY (Maintenance Engineer)

- [] Check impeller clearance – max wear ring clearance 2.5 mm. Replace wear ring if clearance exceeds this limit.
- [] Check shaft runout – max 0.05 mm TIR
- [] Coupling alignment check – max 0.10 mm offset, 0.05 mm angular
- [] Check all gaskets and O-rings for condition

7.5 ANNUAL (OEM Service recommended)

- [] Complete pump overhaul – disassembly, inspection, reassembly
- [] Replace all bearings (regardless of condition) if >8000 hours run
- [] Replace mechanical seal if >12 months continuous operation
- [] Hydraulic performance test – compare with factory test curve

SECTION 9 – TROUBLESHOOTING

Problem 1: HIGH BEARING TEMPERATURE

Possible Causes:

- a) Insufficient lubrication – Check interval vs Section 6.3
- b) Over-greasing – Verify quantity per Section 6.2
- c) Wrong grease type – Verify grease specification
- d) Bearing damage – Check vibration, listen for noise
- e) Misalignment – Check coupling alignment per 7.4
- f) Overloading – Check motor current vs nameplate

Action: If temperature exceeds 75°C – STOP PUMP.
Do not restart without investigation.

Problem 2: HIGH VIBRATION

Possible Causes:

- a) Impeller damage or partial blockage
- b) Bearing failure
- c) Cavitation (suction pressure too low)
- d) Coupling imbalance or wear
- e) Resonance – check baseplate and piping supports

Action: Vibration >6 mm/s – STOP PUMP IMMEDIATELY.

Problem 3: LOSS OF FLOW / LOW PRESSURE

Possible Causes:

- a) Air entrainment in suction
- b) Worn impeller or wear rings (check clearance)
- c) Partially closed suction valve
- d) Pump rotating in wrong direction
- e) Speed too low (check motor frequency)

Problem 4: MECHANICAL SEAL LEAKAGE

Possible Causes:

- a) Dry running – ensure quench water always flowing during pump operation
- b) Worn seal faces – replace seal
- c) Incorrect installation – check as per seal drawing

WARNING: If seal leaks more than 60 ml/hour – STOP PUMP.

Seal failure can lead to fire if product is flammable.

SECTION 10 – SPARE PARTS LIST (Recommended)

Part Description	Part Number	Qty Recommended
Drive End Bearing	SKF 6309	2 per pump
Non-Drive End Bearing	SKF 6307	2 per pump
Mechanical Seal (complete)	Flowserve FS-28	1 per pump
Impeller Wear Ring (pair)	KBL-KDS450-WR	1 set per pump
Shaft Sleeve	KBL-KDS450-SS	1 per pump
Casing Gasket Set	KBL-KDS450-GS	2 sets per pump
Coupling Insert	KBL-KDS450-CI	2 per pump

For technical support: Kirloskar Brothers Ltd.

Regional Office – Mumbai: +91-22-4019-8800

Email: service.west@kirloskarbros.com

END OF EXCERPT